

Buckeye Experimental Aeronautics (BExA) is being built as a dual aerospace engineering program and a student project team. Because of this dual identity, the team is best supported by a complementary relationship between the Aerospace Research Center (ARC) and the Center for Automotive Research (CAR). ARC and CAR offer different strengths that, together, provide the technical depth, operating environment, and accessibility needed for BExA to grow into a long-term student engineering program.

CAR ROLE

- Front-facing student project home
- Late-night student work access
- Project-team friendly environment for assembly and integration
- Member training, meetings, and internal design reviews
- Access to PACE Lab, machine shop, classrooms, and conference rooms
- Hardware storage and day to day operations

ARC ROLE

- Aerospace research alignment
- Faculty mentorship and technical oversight
- High speed aerodynamics and propulsion guidance
- Flight test planning and risk assessment
- Wind tunnel testing and experimental validation
- Graduate / undergraduate research opportunities

WHY CAR MATTERS FOR LONGEVITY

Long term sustainability depends on accessibility and inclusion. Many students, especially underclassmen who don't have cars, rely on convenient, late-access facilities close to campus. CAR's student operated transit system, welcoming project team culture, and facility resources make it the natural gateway for new members. By providing a supportive environment for recruiting, training, and retaining underclassmen, CAR helps build a strong pipeline and ensures the achieves year-to-year continuity.

ARC + CAR MODEL

Our proposed model combines the strengths of both facilities and utilizes them for their intended purposes. ARC provides research alignment, testing guidance, and specialized support that give BExA depth and credibility as an aerospace team. CAR serves as the front-facing student project home, providing the space, tools, and daily operating environment needed for assembly, integration, meetings, member development, documentation, and overall daily project execution tasks. Together, ARC and CAR create a complementary structure that supports both high-quality engineering practices and a thriving student project team.